

Random Samples and Probability Models

Answer Key

Grades 6–7 Mathematics

Quick Answers

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|----------------------------------|-------------------------|
| 1. $\frac{2}{5}$ | 6. $\frac{7}{25}$, 252 |
| 2. 22.5, 20 | 7. $\frac{1}{6}$, 50 |
| 3. $\frac{3}{20}$, 80 | 8. 18, 12 |
| 4. $\frac{3}{5}$, $\frac{2}{5}$ | 9. $\frac{3}{4}$, 600 |
| 5. 2, 24 | 10. — |
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Curriculum

Level: Grades 6–7 Mathematics

6.SP.B.5 – *problems 2, 5, 8*

7.SP.A – *problems 3, 6, 9*

7.SP.C.7 – *problems 1, 4, 7, 10*

Difficulty 1–5 of 5 across 19 tagged checks.

Common wrong answers

8. *If they got 19: reported the range instead of the middle-half spread*

1. Raffle drum: 20 tickets, 8 blue, one drawn.

Solution: Every ticket is the same size and shape, so the 20 outcomes are equally likely and the probability model is favourable over possible. Favourable outcomes: the 8 blue tickets.

$P(\text{blue}) = \frac{8}{20}$, and dividing both parts by 4 puts it in lowest terms.

$\frac{2}{5}$

Sense check: $\frac{2}{5}$ is less than half, and fewer than half the tickets are blue.

2. Homework sample: 15, 20, 25, 30, 20, 35, 20, 15 minutes.

(a) The mean is the total shared equally among the eight students. Total: $15 + 20 + 25 + 30 + 20 + 35 + 20 + 15 = 180$ minutes. Divide by the number of values: $180 \div 8$. 22.5

(b) The mode is the value that appears most often. Counting: 15 appears twice, 25, 30 and 35 once each, and one value appears three times. 20

Reading the pair together: a typical student in this sample worked about 22.5 minutes, and the single most common time was 20 minutes.

3. Tagged fish: 12 tagged and released, sample of 40 contains 6 tagged.

(a) The fraction of the sample that is tagged is $\frac{6}{40}$; dividing both parts by 2 gives lowest terms. $\frac{3}{20}$

(b) A random sample stands in for the pond, so about $\frac{3}{20}$ of all the fish should be tagged. Exactly 12 fish are tagged, and 12 has to be $\frac{3}{20}$ of the pond. Writing that as $\frac{12}{N} = \frac{6}{40}$ and cross-multiplying gives $6N = 12 \times 40 = 480$, so $N = 480 \div 6$. 80
Sense check: $\frac{3}{20}$ of 80 is 12 ✓. Estimate, not a count — another netting would give a slightly different number.

4. Prize spinner: five equal sections — two red, one blue, two green.

(a) The five sections are equal, so the five outcomes are equally likely. “Not red” happens on the blue section and the two green ones, which is three of the five sections. $\frac{3}{5}$

(b) Green covers two of the five sections. $\frac{2}{5}$

Check: $P(\text{red}) = \frac{2}{5}$, and $\frac{2}{5} + \frac{3}{5} = 1$ — an event and its complement always total one whole.

5. Pets per household: 0, 0, 1, 1, 1, 2, 2, 2, 3, 3, 4, 5.

(a) The values are already in order, and there are twelve of them, so the median sits between the sixth and the seventh. Both of those values are 2, so the median is their average. 2

(b) The total is the sum of all twelve values: $0 + 0 + 1 + 1 + 1 + 2 + 2 + 2 + 3 + 3 + 4 + 5$. Group them: $(1 + 1 + 1) = 3$, $(2 + 2 + 2) = 6$, $(3 + 3) = 6$, then $3 + 6 + 6 + 4 + 5$. 24

6. Walkers: random sample of 50 from 900 students, 14 walk.

(a) The sample fraction is $\frac{14}{50}$; dividing both parts by 2 gives lowest terms. $\frac{7}{25}$

(b) The sample was drawn at random from the enrolment list, so it represents the school: apply the same fraction to all 900 students. $\frac{7}{25} \times 900 = 7 \times 36$. 252

Sense check: $\frac{7}{25}$ is a little more than a quarter, and a quarter of 900 is 225, so 252 is the right size.

7. Number cube: fair cube, one roll, then 300 rolls.

(a) A fair cube has six equally likely faces and exactly one of them shows a four. $\frac{1}{6}$

(b) In the long run each face turns up about $\frac{1}{6}$ of the time, so the predicted count is the probability times the number of rolls: $\frac{1}{6} \times 300$. 50

Say it carefully: this is a prediction, not a guarantee. A real run of 300 rolls landing on four somewhere near fifty times is exactly what the model expects.

8. April temperatures: 12, 15, 15, 18, 22, 27, 31 degrees Fahrenheit.

(a) Seven values in order, so the median is the fourth one — three below it and three above it. 18

(b) The interquartile range measures the middle half. Split the sample at the median. Lower half 12, 15, 15 has median $Q_1 = 15$; upper half 22, 27, 31 has median $Q_3 = 27$. Then $\text{IQR} = 27 - 15$. 12

The classmate’s 19 is the *range*: $31 - 12$, the spread from the coldest day to the warmest. The range uses only the two most extreme days, so one unusual day moves it; the IQR describes the middle half and ignores both tails.

9. Skate park survey: 60 asked at the skate park, 45 say skateboarding; school has 800 students.

(a) $\frac{45}{60}$, and dividing both parts by 15 gives lowest terms.

$\frac{3}{4}$

(b) Applying that fraction to the whole school gives $\frac{3}{4} \times 800$.

600

(c) (Open response — judge by reading.) Model answer: the sample was not random. Students at a skate park after school are far more likely to prefer skateboarding than students in general, so the sample is biased toward the answer being measured and 600 is almost certainly too high. The arithmetic in (b) is right; the sample it rests on is not.

A trustworthy method: number all 800 students on the enrolment list and use a random number generator to choose 60 of them, then ask those students — every student has the same chance of being picked, whatever they do after school.

Look for: the student names the bias (where and who was sampled) rather than only saying “too few people”, and describes a selection method that gives every student an equal chance.

10. Simulation design. (Open response — judge by reading.)

Model answer: the four prize cards are equally likely, so I need a device with four equally likely outcomes. Use a spinner with four equal sections, or a random number generator producing the digits 1 to 4, or a bag with four identical slips numbered 1 to 4.

Let outcome 1 stand for the card I want and 2, 3, 4 stand for the other three cards. One trial is one spin, one generated digit, or one slip drawn and put back — one trial models buying one box.

Run at least 100 trials, recording each result, then estimate the probability as the number of trials that came out 1 divided by the number of trials. With many trials that fraction should settle near $\frac{1}{4}$, the theoretical probability.

Look for: a device with four equally likely outcomes, an explicit rule linking outcomes to the event, one trial clearly defined as one box, replacement (the card in the next box does not depend on the last), and a plan to divide successes by total trials.